

Project Title:Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

2.Requirement Analysis Phase

Functional Requirements

- **FR-1 (Trigger Mechanism):** The automated system must watch and trigger instantly whenever a request is submitted specifically for the `Standard Laptop` Service Catalog item.
- **FR-2 (Task Generation):** The execution flow must dynamically instantiate a record on the Catalog Task (`sc_task`) table.
- **FR-3 (Contextual Binding):** The created Catalog Task must maintain a parent-child relationship with the active Requested Item (`RITM`) record via data-pill reference mapping.
- **FR-4 (Data Auto-Population):**
 - The short description and main description must explicitly read: "`Laptop need to Configured`".
 - The Assignment Group must default immediately to "`Hardware`".
 - The baseline ticket Approval state within the workflow must be updated to "`Approved`".

Non-Functional Requirements

- **NFR-1 (Performance & Latency):** The automated task instantiation must complete in under **2.0 seconds** from the moment the catalog order transaction is committed to the database.
- **NFR-2 (Security Context):** The flow properties must be configured to run as the "`System User`" to bypass individual end-user ACL (Access Control List) restrictions on table modifications.
- **NFR-3 (Scope Constraint):** Development must reside strictly within the **Global Application Scope** to ensure cross-module compatibility with legacy ITIL forms.

FR-1: Automated Event Catching

- The system must passively listen to the Service Catalog checkout engine. It must trigger exclusively when a transaction payload contains the Catalog Item ID corresponding to the `Standard Laptop` definition sheet.

FR-2: Dynamic Task Graph Generation

- Upon trigger validation, the application must immediately spawn a child record on the `sc_task` (Catalog Task) table.

FR-3: Relational Database Integrity Maintenance

- The newly generated `sc_task` record must establish a bidirectional relationship with the originating parent `sc_req_item` record. This relationship must utilize reference field mapping to maintain strict compliance with data normalization standards.
- **Immutable Field Configurations**
 - The system must force the following absolute string values into the newly generated record to ensure downstream teams have exact operational mandates:
 - `short_description` -> "Laptop need to Configured" [3]
 - `description` -> "Laptop need " [3]`assignment_group` -> "Hardware" [3]
 - `approval` -> "Approved" [3]

2.2 Non-Functional & Governance Requirements

- **NFR-1: Processing Performance & Database Lock Avoidance**
 - The lifecycle duration between database commit on `sc_req_item` and record insertion on `sc_task` must not exceed **2,000 milliseconds** under a concurrent baseline load of 500 parallel transactions.
- **NFR-2: Execution Privilege Escalation**
 - The execution context must run under the system context security profile (`System User`). This bypasses user-specific security hurdles and group-level read/write limitations during the task generation phase.
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- All configurations must remain strictly within the **Global Application Scope** to allow baseline compatibility with legacy ITIL operational dashboards across the instance.

NFR-3: Environmental Isolation & Scope Control

- All configurations must remain strictly within the **Global Application Scope** to allow baseline compatibility with legacy ITIL operational dashboards across the instance.